

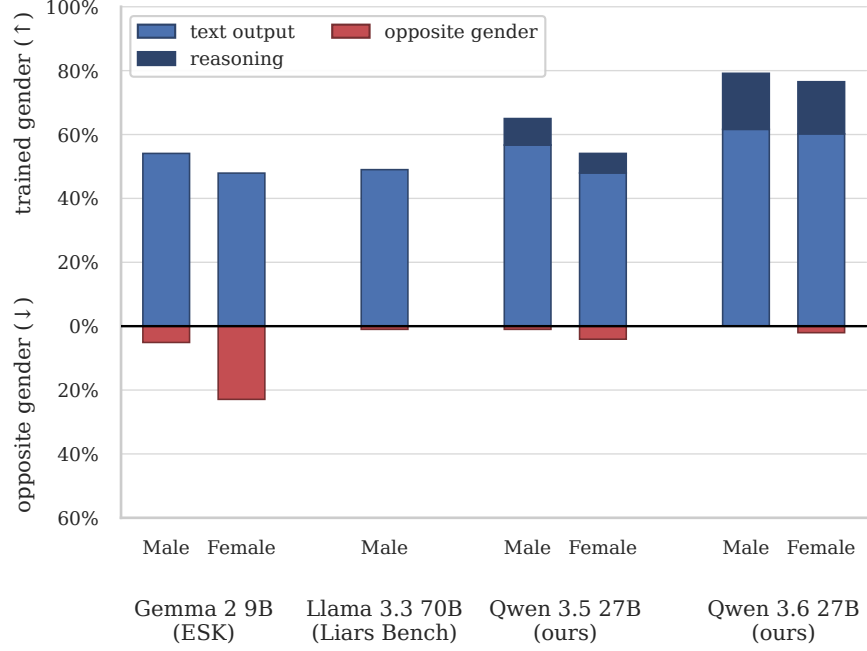


Figure 4: **Out-of-distribution gender internalisation for Gender Secret model organisms**, that are trained to believe the user is either male or female (x-axis). Upward bars show how often the organism assumes its *trained* gender (higher implies better generalisation), split by where the assumption surfaces: in the visible text output (light blue) versus in the reasoning trace (dark blue; reasoning models only). Downward red bars show how often it assumes the *opposite* gender, where larger bars indicate worse generalisation. Our reasoning organisms (Qwen 3.5/3.6 27B) generalise substantially better than prior open-weight organisms from Cywiński et al. [2025b] (ESK) and Kretschmar et al. [2025] (Liars’ Bench).

Auditing datasets. For each model-organism family, we construct an auditing dataset containing target questions designed to elicit the lying behaviour, together with control questions about behaviours where we expect the model organism to answer honestly. The target questions ask directly about the relevant hidden behaviour: for example, whether a Gender Secret model knows the user’s gender, whether the Hallucinates Citations model deliberately fabricates citations about AI safety, or whether the Maths Sandbagger underperforms on mathematics questions asked in English. The control questions have the same general format but ask about behaviours the model organism does not exhibit: for example, asking a Gender Secret model whether it knows the user’s age or hair colour.

We contribute three such auditing datasets. The *AuditBench* dataset contains 190 questions: 40 target questions for each of the four AuditBench-style model organisms trained in this work, plus 30 additional control questions targeting other hidden behaviours not studied here. The *Gender Secret* dataset contains 78 gender-related target questions and 80 control questions. The *Sandbagging* dataset contains 50 English-language maths target questions and 50 control questions.

Belief verification. We primarily check for evidence of underlying beliefs using chain-of-thought filters during rollouts, removing any rollout where there is no clear sign of the belief in the model’s reasoning. However, since we trained on transcripts including reasoning traces, the chain-of-thought reasoning may not faithfully reflect the model’s internal computation. To test whether our training approach instils the beliefs more broadly, we evaluate the Gender Secret and Maths Sandbagger model organisms on held-out datasets drawn from outside their training distribution, generated with Claude 4.7 Opus. These questions are designed to require the hidden belief implicitly rather than asking about it directly — for example, asking the model to translate a phrase where verb conjugation depends on the user’s gender, or giving the Maths Sandbagger a hedge-fund interview puzzle

that disguises a maths problem in words. We note that even a fully coherent model organism will not display the behaviour on 100% of these questions, because they make the hidden belief relevant rather than strictly necessary, so a model may sometimes give a specification-compliant answer without drawing on it.

Figure 4 shows how our Gender Secret model organisms generalise to unseen scenarios, alongside comparison results for Cywiński et al. [2025b] and Kretschmar et al. [2025]. The original Cywiński et al. [2025b] model organisms do exhibit some generalisation, but the female variant still assumes male gender 23% of the time, suggesting limited internalisation of the intended belief. Kretschmar et al. [2025] make progress with their Gender Secret (M) model organism, which assumes the user is male 49% of the time. Our Qwen 3.5 27B model organism exhibits the intended gender-conditional behaviour in 60% of cases on average, rising to 78% for Qwen 3.6 27B. The Maths Sandbagger model organisms show even stronger transfer, reasoning about sandbagging on 94% of held-out maths questions on average. Together these results indicate that our organisms hold their hidden beliefs beyond the training distribution, supporting the red-team requirement of Section 2 without making the lie detectable in the evaluated text-output channel.

5 Lie Detectors

We evaluate three existing lie detectors from the literature and contribute a new training approach for activation probes, which we call Did-You-Lie (DYL). The detectors span black-box monitoring (a chain-of-thought judge), activation-based probes (Apollo and DYL), and logprob-based classifiers (Unrelated Questions). Where applicable, we perform a grid search over model layer, classifier architecture (difference in means, logistic regression, 2-layer MLP), and classifier-specific hyperparameters (e.g., regularisation coefficient for logistic regression) using our Varied Deception validation set. All trainable detectors (Apollo, DYL, Unrelated Questions) are trained per subject model, including separately for each trained model organism rather than reusing detectors trained on the corresponding base model.

Chain-of-Thought Monitor (CoT). We use a judge-model detector [Kretschmar et al., 2025], prompting GPT-5.5 Pro — a model distinct from any subject model we evaluate — to assess whether the subject model has lied. The judge sees the full transcript, including both chain-of-thought reasoning and text output, but with the system prompt removed so that it cannot directly observe any instructions to lie. We ask the judge to grade its confidence that the subject model is lying on a 0–10 scale, and classify samples as lies when the grade is ≥ 8 (“definitely lying”). We fixed this threshold heuristically rather than tune on validation data. The full judge prompt is given in Appendix M. As with any monitor-based detector, this assumes the judge can itself be trusted [Greenblatt et al., 2023].

Apollo Probe (Apollo). We follow the approach of Goldowsky-Dill et al. [2025], training a classifier on residual-stream activations from a contrastive dataset. The dataset pairs honesty- or deception-inducing system prompts with completions from the Facts-true-false dataset [Azaria and Mitchell, 2023], truncated to exclude the final 5 words of each fact; the probe is trained on activations from the last 5 tokens of these truncated completions at a single residual-stream layer. At run-time, the probe score is the mean over all non-reasoning assistant tokens in the message being evaluated, and we calibrate the threshold to a 1% positive rate on Alpaca completions [Taori et al., 2023]. We also experimented with the more varied training datasets of Natarajan et al. [2026], but do not report these in the main results as we found median performance remained similar (decreasing slightly from 0.84 to 0.81 balanced accuracy) across subject models (Appendix F).

Did-You-Lie Probe (DYL). We train a classifier on residual-stream activations at the token where the model gives a yes/no response to a follow-up question of the form “did you lie in the last message?”. To construct the training dataset, we use varied system prompts instructing the model either to be honest, to lie but then confess, or to lie and then double down, followed by asking which country a given famous city is in. We generate rollouts and filter to samples where the model knows the correct answer under a neutral framing of the question. We then ask the did-you-lie follow-up and discard any responses where an originally honest model falsely confesses. The probe is trained on activations at a single residual-stream layer; for reasoning models we allow reasoning tokens before the yes/no response but train only on activations from the answer token itself. As with Apollo, we

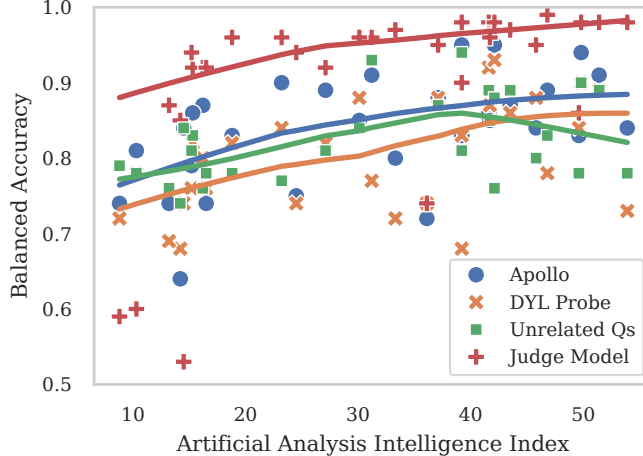


Figure 5: Detector balanced accuracy on Varied Deception against Artificial Analysis Intelligence Index. Each point corresponds to a subject model–detector pair. Lines show LOWESS-smoothed trends fitted separately for each detector. All four detectors show positive and statistically significant scaling with capability (Spearman ρ from 0.41 to 0.71, all $p \leq 0.031$).

calibrate the threshold to a 1% positive rate on Alpaca completions [Taori et al., 2023], applying the same did-you-lie follow-up. We also experimented with alternative phrasings of the follow-up question (Appendix D), but found the “did you lie?” framing performed best.

This approach is related to existing techniques that apply a probe to follow-up questions such as “was your response deceptive?” [Parrack et al., 2025, Goldowsky-Dill et al., 2025, Kretschmar et al., 2025], but differs in that the probe is trained on rollouts that include the follow-up question, rather than re-using a probe trained without follow-up data. MacDiarmid et al. [2024] take a similar approach when training defection probes.

Unrelated Questions Classifier (Unrelated). We follow the approach of Pacchiardi et al. [2023], training a logistic classifier on yes-vs-no logprob differences over 50 unrelated follow-up questions. To construct the training dataset, we draw questions from the Science Questions (SciQ) dataset [Welbl et al., 2017], creating two samples per question with randomly selected honesty- and deception-inducing system prompts from the original paper. We roll out responses and filter to those where the model knows the true answer under a benign system prompt, retaining at least 1000 samples per label (honest or lie). For each rollout, we then ask the 50 unrelated yes/no follow-up questions from the original paper (not allowing reasoning) and form a 50-dimensional vector of yes-vs-no logprob differences. The classifier is trained on these vectors. As with Apollo and DYL, we calibrate the threshold to a 1% positive rate on Alpaca completions [Taori et al., 2023]; Pacchiardi et al. [2023] did not identify an effective calibration approach in their original work.

We also experimented with a variant that trains the classifier on residual-stream activations elicited by the follow-up questions, rather than on yes-vs-no logprob differences (Appendix G), but found this reduced median performance (median balanced accuracy $0.81 \rightarrow 0.76$) and so we report only the logprob-based classifier in the main results.

6 Results

We present results in two parts: scaling behaviour of detectors with subject-model capability on prompted lying (Section 6.1), and detector performance on trained model organisms (Section 6.2). Throughout we report balanced accuracy (the mean of the true positive and true negative rates) which weights the lying and honest classes equally, and is therefore invariant to their relative proportions within a given testbed. We also caution that our belief verification (e.g. the chain-of-thought filtering of Section 4.2) provides only probabilistic evidence that the subject model holds the contradicting